A Systematic Review on the Ethical Implications of AI-Generated Contents: Trust, Deception, and Public Sentiment

Abstract

The rapid proliferation of AI-generated media, such as deepfakes and synthetic content, has significantly transformed content creation while introducing critical ethical challenges, including erosion of trust, misinformation, and potential risks to public sentiment. Despite numerous studies addressing these issues, existing research often lacks integrated ethical frameworks that encompass transparency, accountability, and cultural differences in public trust, with limited attention to long-term societal impacts. This study employs a systematic literature review and content analysis of 72 peer-reviewed articles (2015–2025), selected from an initial pool of 724, to address gaps in understanding trust dynamics, technological vulnerabilities enabling deception, and regional disparities in the perception of synthetic media. The research design focuses on investigating mechanisms of trust erosion, socio-technical drivers of AI-induced deception, and cultural-demographic influences on public perceptions. The study proposes an AI-Driven Trust Resilience Framework that integrates detection technologies, policy reforms, and educational initiatives. The study highlights the need for adaptive, cross-cultural, and ethical frameworks to safeguard digital content integrity and restore public trust, offering insights into detection system limitations and providing actionable strategies to mitigate the adverse effects of AI-driven deception.

Keywords: AI-generated media; misinformation; synthetic content; digital media ethics; public trust; AI deception

1 Introduction

The rapid integration of artificial intelligence (AI) into media has revolutionized the content of creation, distribution, and personalization, enabling unprecedented efficiency and creativity. Technologies such as Generative Adversarial Networks (GANs) and diffusion models now produce synthetic media—images, audio, and videos that are nearly indistinguishable from authentic content, thereby transforming industries such as entertainment, advertising and journalism [1, 2]. However, this progress has introduced significant ethical challenges, particularly through the proliferation of deepfakes:

intentional deception that manipulates public perception by falsifying events or impersonating individuals [3]. Studies have revealed that 96% of deepfakes are maliciously deployed, including political propaganda and non-consensual pornography, and generative text tools such as ChatGPT have been found to exacerbate misinformation on a large scale. These tools can unintentionally amplify false or misleading narratives [4].

The rise of AI-generated media threatens foundational trust in the information ecosystem. Platforms such as Facebook and TikTok employ AI detection tools to combat synthetic content [5]; however these systems struggle to keep pace with advancing generative technologies. This creates a paradox: while AI enhances creative potential, it simultaneously erodes content credibility, enabling misinformation campaigns that destabilize democratic processes and public discourse [6, 7] . For instance, AI-generated text (ChatGPT) and visuals (DALL-E 3) [8] . The consequences extend beyond individual harm, undermining societal cohesion and institutional trust, as seen in recent cases of election interference. Despite growing awareness, critical research gaps persist. Existing studies lack integrated frameworks to address ethical concerns such as bias, transparency, and accountability in AI-generated content [9]. Few studies have explored regional and cultural variations in public trust or skepticism towards synthetic media, thus limiting global applicability [6]. Long-term societal and economic impacts remain under examination, and mitigation strategies that combine policy, education, and technology are scarce. In addition, the mechanisms that rebuild trust after exposure to deceptive AI content are poorly understood .

This study aims to overcome these limitations by synthesizing interdisciplinary insights from the literature published between 2015 and 2025 to propose actionable strategies.Through a systematic literature review and content analysis, this study proposed strategies to mitigate the risks associated with AI-generated content. Specifically, it seeks to (1) evaluate the impact of AI-generated content on trust within media ecosystems, (2) analyze the technological and societal factors that contribute to the rise of deceptive practices, such as deepfakes, and (3) assess cultural and demographic differences in public responses to synthetic media. The key contributions of this study are as follows:

- **Comprehensive Systematic Literature Review:** Our study provides a comprehensive review of the ethical challenges posed by AI-generated content, spanning trust erosion in media ecosystems, the proliferation of misinformation, and risks to democratic integrity across fields such as media ethics, digital trust, and political communication.
- **Content Analysis Insights:** Identifying technological vulnerabilities such as the limitations of deepfake detection tools and evaluating sociopolitical interventions, including regulatory frameworks and transparency protocols.
- **Multidisciplinary Framework Proposal:** The proposed framework integrates technological, regulatory, educational, and cultural dimensions to ensure ethical AI deployment, rebuild trust, and mitigate the possible risks. This study advocates for policy reforms that include stricter content labelling laws and a comprehensive strategy to restore trust and accountability in AI-mediated media ecosystems.

The paper is organized as follows: Section 2 provides a review of the literature on AI media technologies, societal risks, and policy responses to deceptive content. Section 3 outlines the methodology, detailing the study design, article selection process, and data extraction procedures. Section 4 presents an analysis of bibliometric trends, including publication patterns, geographical distribution, and institutional contributions. Section 5 focuses on content analysis, examining the erosion of trust in media ecosystems, transparency in AI deployment, and the economic and political implications of deceptive mechanisms. Section 6 explores public sentiment and societal perspectives on the credibility of AI-generated content, cultural and demographic variations, and associated risks and mitigation strategies. Section 7 offers a comparative analysis of global perspectives on trust and misinformation, supported by case studies. Section 8 addresses ethical implications, the role of AI developers, and provides policy recommendations for AI governance. Section 9 proposes a three-stage framework, emphasizing technological solutions, regulatory interventions, and educational initiatives. Section 10 discusses the challenges and limitations faced in methodology, technology, and sociopolitical barriers. Section 11 outlines future research directions, including ethical integration, the enhancement of detection systems, and the exploration of regulatory strategies. Section 12 concludes with key insights and actionable solutions for the responsible management of AI-generated content.

2 Related Works

The integration of AI across diverse sectors such as education, media, healthcare, business, and governance has not only fueled creativity and productivity but has also sparked significant scholarly discussions on its ethical implications and long-term societal consequences. This section presents the most relevant recent studies on these issues.

Ranade and Saravia [10] conducted a systematic literature review (SLR) on teaching AI Ethics in Technical and Professional Communication and proposed a three-tier framework to align institutional, course, and instructional objectives. The study highlights the gap between theoretical ethical principles and practical implementation in education. Contreras and Jaimes [11] expanded this discourse through a systematic literature review (SLR) focused on AI ethics in education and research, identifying transparency, privacy, and equity as dominant yet inadequately addressed themes. Their key findings highlight a persistent gap between the theoretical acknowledgment of these ethical principles and their fragmented operationalization. Specifically, transparency is frequently limited to superficial disclosures regarding AI usage, neglecting deeper accountability in algorithmic decision-making. Privacy measures remain inconsistently applied, and approaches to equity often lack coherent strategies, perpetuating existing inequalities. Xu et al. [12] further contributed to this discourse by examining the role of large language models (LLMs) within personalized learning contexts. Their research emphasized that while LLMs have significant potential to enhance accessibility and inclusivity, they simultaneously present considerable ethical challenges, particularly related to algorithmic bias and data privacy breaches. Collectively,

Table 1: Summary of Studies: Methods, Key Findings, and Limitations

Ref.	Key Findings	Limitations
Ranade and Saravia [10]	Proposed a three-level framework (institutional, course, instruction) to integrate AI ethics literacy into technical communication curricula.	Limited focus on non-Western educational contexts; lacked empirical validation of framework effectiveness.
Contreras and Jaimes [11]	Identified transparency, privacy, accountability, and equity as dominant ethical themes in AI for education/research.	Excluded non-English studies; limited analysis of cultural variations in ethical priorities.
Xu et al. [12]	Demonstrated AI and LLMs enhance personalized learning and accessibility but pose risks like algorithmic bias and data privacy breaches.	Narrow database scope (IEEE, ACM); lacked sector-specific mitigation strategies.
Gunawan and Wiputra [13]	Highlighted ethical concerns in generative AI for medicine: patient autonomy, clinical bias, and privacy.	Limited empirical validation; focused on theoretical frameworks over practical implementation.
Masciari et al. [14]	Found transformer models (BERT) improve recommended system accuracy but neglect ethical issues like fairness and explainability.	Overemphasis on technical solutions; insufficient sociopolitical analysis (policy gaps).
Bena et al. [15]	Proposed Big Data Governance Maturity Model (BDG MAM) to address data security, scalability, and compliance challenges in intelligent systems.	Limited sector-specific analysis (healthcare vs. finance); under emphasized public education.
Han et al. [16]	Showcased GPT-based LLMs automate business tasks but risk misinformation and accountability gaps.	Preliminary findings; excluded non-academic sources (industry reports).
Abokhoza et al. [17]	Emphasized need for ethical frameworks in AI-driven communication governance to prevent privacy violations and manipulation.	Conceptual focus; lacked case studies on real-world policy implementation.
Wiese et al. [18]	suggested proactive approaches to responsible AI through Quintuple Helix model (academia, government, industry). Highlighted gaps in translating principles to practice.	Relied on niche databases (PhilPapers).
Koralage et al. [19]	Explored AI's role in accounting education (intelligent tutors) but noted gaps in interdisciplinary ethical frameworks.	Narrow focus on accounting; lacked cross-sector comparisons.

these studies underscore the critical need for pedagogical strategies that carefully balance technological innovation with ethical accountability, aligning closely with the transparency, privacy, and equity concerns highlighted by Contreras and Jaimes [11].

Existing literature on AI ethics has predominantly focused on sector-specific challenges, often overlooking broader cultural and regional disparities (Table 1). For instance, studies by Han et al. [16] and Masciari et al. [14] primarily addressed Western contexts, neglecting variations in trust dynamics arising from different global regulatory environments. Additionally, Koralage et al. [19] highlighted a critical gap in interdisciplinary approaches within accounting pedagogy. The current literature largely emphasizes technical solutions while policy and educational interventions remain isolated and insufficiently integrated. Methodological limitations have exacerbated these issues. Wiese et al. [18] overly relied on niche databases such as PhilPapers, limiting their exposure to diverse perspectives, whereas Contreras and Jaimes [11] excluded non-English literature, thus skewing their findings toward anglophone contexts.

Gunawan and Wiputra [13] reviewed the ethical challenges associated with generative AI and reveal significant concerns regarding patient autonomy, biased clinical decision-making, and data privacy. Their findings underscore the necessity for transparent AI systems that uphold patient rights and comply with regulatory standards. Similarly, Masciari et al. [14] analyzed AI-driven recommendation systems beyond the medical field, noting advancements in transformer models such as BERT that enhance recommendation accuracy. However, their review criticized the neglect of ethical considerations, specifically algorithmic fairness, highlighting how biased recommendations may perpetuate inequities. This aligns with broader critiques of data governance articulated by Bena et al. [15], who investigated the ethical challenges posed by intelligent systems (IoT and ML) concerning data security and scalability. Their Big Data Governance Maturity Assessment Model (BDG MAM) provides a structured road-map while highlighting unresolved ethical issues in automated decision-making.

Han et al. [16] also reviewed GPT-based large language models in economics, demonstrating their utility in automating tasks such as market analysis but also flagging risks such as misinformation propagation and accountability gaps, particularly when AI-generated content influences financial decisions without human oversight. Extending these critiques to communication governance, Abokhoza et al. [17] argued for stringent ethical frameworks to manage AI's role in regulating information flows, emphasizing the prevention of privacy violations and manipulative practices. Cross-sectoral research by Wiese et al. [18] adopted the Quintuple Helix model to analyze proactive approaches to responsible AI innovation across academia, government, and industry, further highlighting the gap between high-level ethical principles and their practical implementation—a recurring theme across various sectors.

To address these shortcomings comprehensively, this study synthesizes insights from broader, more interdisciplinary sources such as Google Scholar and IEEE databases, which are renowned for extensive coverage and rigorous standards. This approach enriches the pedagogical framework by integrating IEEE guidelines on algorithmic transparency and global perspectives on data ethics, as discussed by Bena et al. [15] and Ranade and Saravia [10]. By bridging the technical, cultural, and policy dimensions, our analysis offers comprehensive strategies exemplified by the AI-Mediated Trust Resilience Framework, emphasizing equity, transparency, accountability, and public empowerment.

3 Methodology

In this section, we outline our study design by detailing the research questions formulated to address existing gaps, the systematic literature search strategies implemented, the rigorous article selection process, methods employed for bias mitigation, and the use of the PRISMA diagram to visually represent the article selection workflow in Figure 1.

3.1 Study Design

This study uses a systematic literature review (SLR) approach guided by PRISMA guidelines to examine the ethical implications of AI-generated content from an interdisciplinary perspective. The methodology prioritizes transparency, consistency, and rigor, focusing on formulating clear research questions, designing robust search strategies, applying strict inclusion/exclusion criteria, and documenting the selection process using a PRISMA flow diagram. To address gaps in existing literature, the review is structured three research questions:

- **RQ1:** How does AI-generated content influence trust dynamics within media ecosystems, and what mechanisms exacerbate or mitigate trust erosion?
- **RQ2:** What technological vulnerabilities and sociocultural drivers enable AI-driven deception?
- **RQ3:** How do cultural, demographic, and regional differences shape public perception and skepticism toward synthetic media?

3.2 Search Strategy

The search string is designed to capture interdisciplinary studies that examine the ethical, technical, and sociocultural dimensions of AI-generated content. It employs Boolean operators to combine three thematic clusters. The first cluster pertains to technical terms associated with AI technologies and content creation tools, including "AI-generated content," "synthetic media," "deepfakes," "generative AI," "GANs," "LLMs," and "large language models." The second cluster addresses ethical considerations, with keywords such as "trust erosion," "misinformation," "disinformation," "algorithmic bias," "ethical frameworks," and "privacy violations," which focus on governance, risks, and ethical implications. The third cluster explores sociocultural factors, using terms like "cultural differences," "policy interventions," "detection systems," "public perception," "media literacy," and "regulatory compliance" to investigate societal impacts and responses to AI-generated content. To ensure the inclusion of only peer-reviewed, scholarly works, the search excludes non-peer-reviewed document types such as editorials and letters. The full search string utilized is as follows:

```
(ALL=("AI-generated content" OR "synthetic media" OR "deepfakes" OR "
    generative AI" OR "GANs" OR "LLMs" OR "large language models")
AND ("trust erosion" OR "misinformation" OR "disinformation" OR "
    algorithmic bias" OR "ethical frameworks" OR "privacy violations")
```

```
AND ("cultural differences" OR "policy interventions" OR "detection
    systems" OR "public perception" OR "media literacy" OR "regulatory
    compliance"))
NOT (DT=("CONFERENCE PAPER" OR "BOOK SECTION" OR "EDITORIAL" OR "
    COMMUNICATION"))
```

Table 2 summarizes the number of articles retrieved from each data source using the specified search string by considering the areas of the papers retrieved across different academic databases, reflecting their relevance to the study's interdisciplinary scope.

Table 2: Research Articles Retrieved by Data Source (2015–2025) and Paper Counts

Data Source	Number of Papers
IEEE Xplore	78
ACM Library	65
SpringerLink	92
Google Scholar	145
ScienceDirect	54
Scopus	102
JSTOR	37
PubMed	29
Wiley Online Library	41
Taylor & Francis Online	48
ProQuest	33
Total Paper Count	724

3.3 Article Selection Procedure

The article selection process followed a rigorous, multi-step protocol designed to ensure methodological transparency, scholarly rigor, and alignment with the study's objectives. Initially, a comprehensive search was conducted across 11 academic databases (as shown in Table 2), which yielded a total of 724 articles. Duplicate entries were systematically identified and removed using automated tools, followed by manual verification, resulting in 650 unique articles (as shown in Figure 1). In the subsequent step, a peer-review screening process was carried out to ensure that only verified journal and conference papers were included, reducing the number of articles to 520. Following this, non-English language publications were excluded, which further narrowed the corpus to 480 articles. This exclusion introduced a geographic bias, which is explicitly recognized as a limitation of this research. A further refinement was made by restricting the corpus to research articles only, excluding other types of publications, such as editorials, letters, and communications. This step resulted in 310 relevant articles. A detailed relevance assessment was performed based on titles, abstracts, and full-text reviews to ensure alignment with the study's core themes. This thorough and methodical approach ultimately led to the inclusion of 72 highly relevant articles, as shown in Figure 1, providing a well-curated and interdisciplinary dataset.

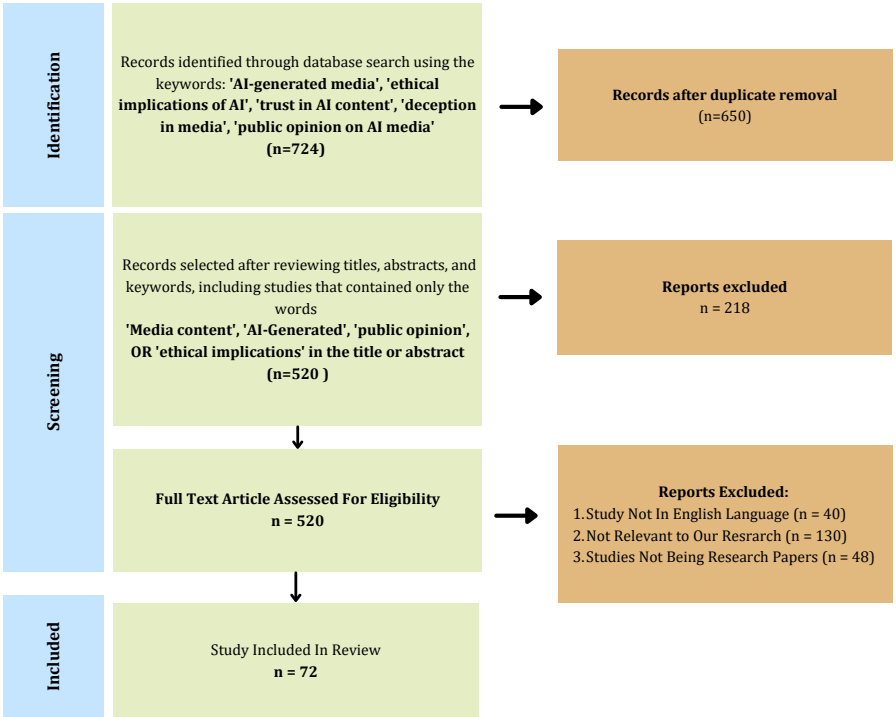

Fig. 1: PRISMA: article selection process, detailing the records identified, screened, and included in the review based on specific keywords and exclusion criteria.

Table 3: Final Article Selection Process and Remaining Papers by Step

Step	Selection Criteria	Remaining	Excluded
Initial Search	11 academic databases	724	-
Duplicate Removal	Automated + manual verification	650	74
Peer-Review Screen	Verified journal/conference papers	520	130
Language Filter	English-language only	480	40
Content Type	Research articles only	310	170
Relevance Assessment	Title/Abstract/Full-text review	72	238

3.4 Bias Mitigation

To minimize bias in the article selection process, our study used a multi-faceted approach that addressed the ethical and socio-technical challenges of AI-generated content. We sourced articles from 11 interdisciplinary databases, such as IEEE Xplore, SpringerLink, and Scopus, ensuring a broad range of global perspectives on deep-fake detection, trust erosion, and cultural differences. Pre-defined exclusion criteria were applied to eliminate non-peer-reviewed sources, non-English articles, and non-scholarly documents, prioritizing objectivity, especially in evaluating sensitive topics like algorithmic bias and policy interventions as presented in Table 3

3.5 Final Selection

The structured, multi-phase selection protocol as per Figure 1, yielded 72 peer-reviewed articles published in IEEE Xplore, Springer, Scopus, and PubMed for comprehensive analysis. These studies, published between 2015 and 2025, met stringent inclusion criteria: scholarly rigor (peer-reviewed), and they are thematically aligned with AI-generated content's societal, technical, and governance dimensions, ensuring their relevance to the field of AI and ethics. The corpus spans diverse domain, such as algorithmic fairness in detection systems. Each article is critically engaged within the review, ensuring methodological and contextual diversity while addressing gaps in existing literature.

4 Bibliometric Analysis

This section presents a bibliometric analysis to systematically explore the evolution and distribution of research on AI-generated content, focusing on its ethical, technical, and sociocultural dimensions. By examining publication trends, key research themes, and geographical patterns, the study identifies the temporal progression and interdisciplinary nature of the field.

4.1 Research Themes and Temporal Progression

The systematic analysis encompassed most relevant 72 high-quality studies, meticulously selected from an initial pool of 724 articles through rigorous screening processes. As illustrated in Table 4, the research delineates a distinct temporal progression: initial studies (2015-2025, 6 studies) concentrated on ethical frameworks and detection prototypes, succeeded by explorations of trust dynamics (2020-2021, 8 studies) and technical advancements in detection systems (2022, 10 studies) . Recent publications (2023-2025, 12 studies) have shifted focus towards cultural disparities, policy interventions, and governance frameworks (2024-2025, 10 studies) , indicative of the field's maturation.

y

4.2 Geographical Distribution

The spatial analysis of research contributions from the selected corpus highlights the distribution of studies across various regions. The data indicates that researchers affiliated with institutions in the United States contributed the highest number of articles (15), followed by researchers from India (12). Significant contributions were also noted from Germany (10), the United Kingdom (8), and Canada (7). Additionally, several studies originated from countries such as Australia (5), China (4), and Brazil (3). This geographic distribution, visualized in Figure 2, provides a clear representation of global academic engagement and the diverse origins of research contributions included in this study.

Table 4: Data Extraction: Year, Journal Ranking, and Key Themes

Year	Journal Ranking	Key Research Themes	Papers Count
2015-2019	Q1-Q3	Ethical frameworks, early detection systems, algorithmic bias	15
2020-2021	Q1-Q2	Trust dynamics in synthetic media, misinformation, disinformation	18
2022	Q1	Detection systems (GANs, LLMs), transparency, bias mitigation	20
2023-2025	Q1-Q2	Cultural disparities, policy interventions, public perception	19
2024-2025	Q1	AI governance, real-world case studies, regulatory compliance	10

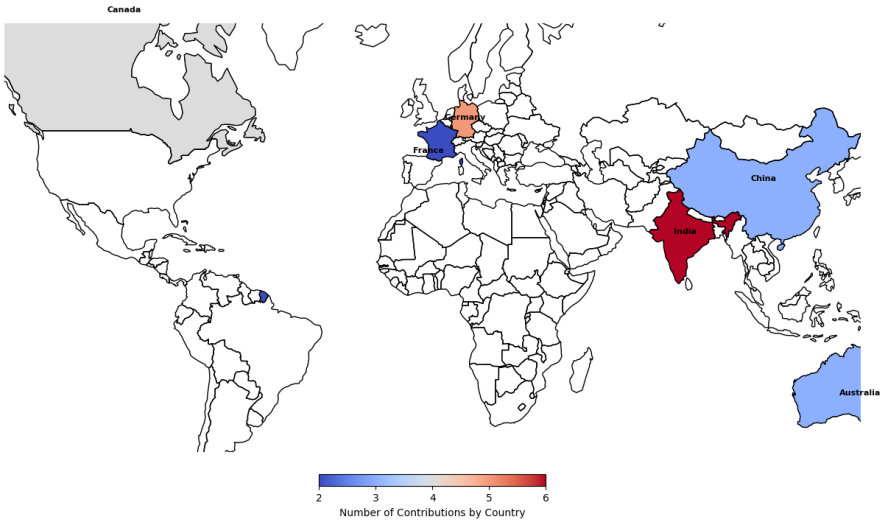

Fig. 2: Geographical Distribution of Research Contributions

4.3 Journal and Publisher Representation

This subsection presents overview of of both journal and publisher representation and annual publication activity. Figure 4 depicts the number of reviewed articles published by each journal or publishing house. Rather than implying any evaluative ranking or contribution, the analysis simply reports the distribution of articles across publishing entities. The visualization facilitates an understanding of the publishing landscape within this research domain and may indicate which outlets have served as key platforms for scholarly discourse on AI-generated content and its ethical, technical,

and societal implications. Figure 3 illustrates the annual distribution of publications from 2021 to 2025 This visual analysis highlights fluctuations in research output across the years, offering insights into changes in academic interest and the potential influence of emerging technologies, societal developments, or funding policies.

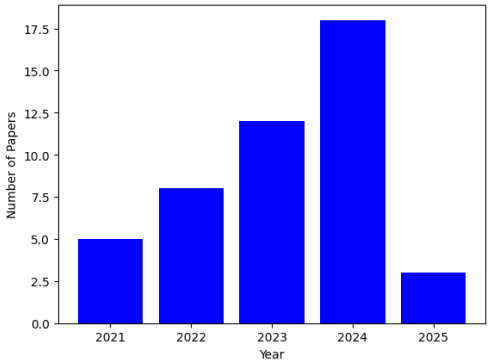

Fig. 3: Publication Trend Over Time

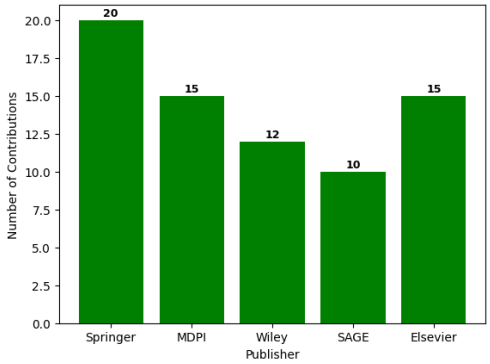

Fig. 4: Publishers on AI Content Ethics

5 Trust and Deception in AI-Generated Contents

Integrating artificial intelligence (AI) tools in content creation has sparked significant accountability concerns, particularly regarding the erosion of trust in media ecosystems As AI-generated content proliferates, the boundary between factual reporting and synthetic media becomes increasingly blurred, leading to widespread skepticism and challenges in maintaining public trust. A survey conducted by Fernandez and Alani [20] pointed out that the disturbing trend that younger adults below 30 have a near equal faith in social media sources of news as they do in national broadcasting channels. The advances in AI have made the boundary between fact and fiction virtually nonexistent, which has, in turn, spawned a trust deficit and fueled a crisis in people's confidence in the media and all other democratic institutions. Nontrustworthy content generation is also rampant in key moments such as elections. According to Krüger and Wilson [21], about 85% of internet users surveyed in 2024 were worried about their ability to distinguish between real and fake online content. The degree of mistrust is also quite widespread of any media content, irrespective of whether it was created by an established media channel or new, owing to distrust of AI-generated content because of its global harm. Jiang et al. [22] points out that, as the public understanding of AI advances progresses, people may come to a generalized skeptical attitude toward journalism, leaving aside stirring checked and rechecked raw data if, say, AI writing tools come to believe to be correct. A summary of different aspects of AI-generated Media is presented in Table 5. The following subsections discusses the erosion of trust, deception, transperancy & accountability, economic and political implications.

Table 5: Aspects of Trust in AI-Generated Media

Study	Aspect	Description	Statistical Data	Implications	Suggested Actions
Krüger and Wilson [21]	Erosion of Trust	Decline due to misinformation	85% worried about fake content	Increased skepticism	Enhance transparency
Fernandez and Alani [20]	Misinformation Impact	Role in public perception	Adults under 30 trust social media as much as national outlets	Erosion of credibility	Implement fact-checking
Newman et al. [23]	Public Engagement	Importance of educating audiences	Higher understanding leads to better trust	Reduces misinformation	Develop educational programs
Jobin et al. [24]	Expert Collaboration	Working with experts for ethical guidelines	Enhanced ethical standards	Greater accountability	Establish partnerships
Fletcher and Nielsen [25]	Community Involvement	Community feedback enhances credibility	Strengthened community ties	Improved perception	Create participatory platforms
Nguyen et al. [26]	Transparency Initiatives	Clear communication about AI use	Transparency boosts credibility	Builds public confidence	Regular updates
Mondal and Mondal [27]	Accountability Mechanisms	Oversight for AI-generated content	Independent audits improve reliability	Reinforces responsibility	Implement oversight committees

5.1 Erosion of Trust in Media Ecosystems

The crisis of dwindling faith in the media sector has become quite dire, incredibly, as the impact of media has been researched extensively in a wide range of reports. In this regard, researchers identify misinformation and disinformation campaigns as contributing factors that reduce the trust barometer. The growth in these posts reflects a specific distrust in mass media; one-third of respondents (36%) claimed to have no trust in mass media Fernandez and Alani [20]. The increased deepfake technology and other AI technologies that create incredibly real yet false material only worsen this. As a result, the audience is prepared to remain skeptical of a news report and is always advised to approach it with caution instead of believing it unquestioningly Krüger and Wilson [21]. According to Brinkerhoff and Brinkerhoff [28], only 18% to 29% of Gen Z in the U.S. trust politicians. In contrast, nearly 70% report that they always fact-check information encountered on social media. For media organizations, the conclusion is unequivocal: they need to change their strategies so that younger audiences can regain their trust, but at the same time, get ready for lower trust in AI-generated content [29], as shown in Figure 5.

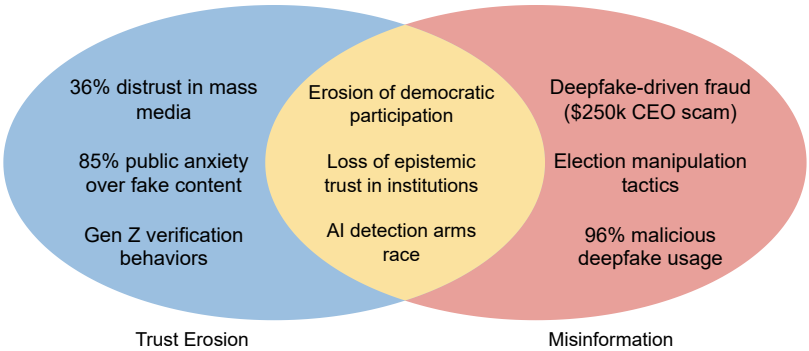

Fig. 5: Decline of trust in media and institutions amid rising misinformation and AI-driven manipulation.

Several measures can be implemented to restore the waning trust resulting from the proliferation of AI-generated content. The most crucial of these is transparency—media institutions must explicitly indicate when content has been created using AI and disclose the algorithms involved. Audiences are generally more accepting of AI-generated content as long as it is clearly labeled, such as in instances of reporting on activities in sectors like finance or weather forecasting [30]. However, significant disagreement persists regarding the standards for appropriate labeling and the extent to which AI usage should be disclosed [23]. To address these concerns, implementing educational programs aimed at audiences could foster a better understanding of AI technologies and their applications. Furthermore, informing the audience about how AI-generated content is produced and the potential benefits it may offer can assist media organizations in cultivating audiences that are more resilient to misinformation [23]. Additionally, collaboration with technology experts and ethicists on the ethical implementation of AI is critical. Content producers can uphold professional standards by developing guidelines that direct the responsible use of AI tools and emphasize the importance of accountability in content creation.

5.2 Deception Mechanisms and Impacts

Among the mechanisms of deception, AI-generated content, particularly deepfake technology, stands out as a significant concern. Deepfakes, powered by Generative Adversarial Networks (GANs), are capable of producing highly realistic synthetic audio, video, and images that depict non-existent events or individuals. While this technology offers transformative potential in fields such as entertainment and education, its misuse has escalated, leading to severe societal consequences. According to Heidari et al. [31], deepfakes have been weaponized for monetary crimes, including impersonation and the fabrication of political speeches or events, which can deceive even astute audiences [32, 33]. This manipulation extends to the political sphere, where, as discussed by Efthymiou and Egleton [34], deepfakes can distort voters perceptions of candidates during elections, fostering distrust and disunity. These dual-use characteristics of deepfake technology highlight its capacity for both innovation

and harm, necessitating robust deterrence measures and counteraction strategies to mitigate its abuse and societal impact.

Table 6: Characteristics of Deepfake Technology.

Study	Aspect	Description	Implications
Heidari et al. [31]	Fraudulent Activities	CEO impersonations to approve fraudulent financial transactions.	Millions of dollars lost worldwide.
Efthymiou and Egleton [34]	Electoral Manipulation	Substituted political addresses employed in elections to duplicitously influence voters.	Doubt in the electoral and democratic process.
Gilbert and Gilbert [32]	Identity Theft	Fabricated videos to steal identities for personal or financial gain.	Significant harm to individual reputations.
Steingartner and Galinec [33]	Strategic Deception	Disinformation campaigns targeting geopolitical adversaries.	Geopolitical instability and misinformation.

5.3 Transparency and Accountability

Accountability and transparency are necessary to restore trust, especially in content created with the help of AI. Diakopoulos [35] emphasizes that organization should explain why AI technologies are used and outline measures to ensure accuracy. Mondal and Mondal [27] suggests that bounding oversight committees or independent audits should be in place to determine the correctness and authenticity of AI-generated content. Furthermore, media organizations should create avenues through which their audiences can provide feedback about the contents delivered. Fletcher and Nielsen [25] emphasize that respecting the set regulations for transparency and accountability help the organizations win the public trust when deploying AI technologies.

5.4 Economic and Political Implications

The misuse of deepfake technology has far-reaching implications across various domains, as highlighted in Table 6, and Table 7. The economic costs of deepfake technology arise from much more than just fraud. They stated that AI-driven disinformation affects negatively in marketplaces, resulting in financial turbulence [34]. For example, impersonation business scams have cost millions. In 2023, an energy company in Europe lost $250,000 when criminals impersonated its CEO using AI-generated audio. Such incidents underscore the potential risks businesses face in the era of AI-driven fraud [36]. Deepfake technology caters as a political threat to the democratic governance configuration. In easier words,some of these risks might be minimized by the decentralized and trustworthy nature of the blockchain used for confirming the media sources [34].

AI-based resources have become increasingly valuable in combating deepfake content by identifying discrepancies in facial movements, voice patterns, and metadata.

Newman et al. [23] proposed a frame-loss-based anomaly detection model for deepfake videos, achieving an average detection rate of 92%. Technological measures such as blockchain verification and embedded proof systems also play a critical role in this defense. Islam et al. [36] demonstrated that blockchain technology enables the verification of the origin of digital content, while AI can invisibly embed markers to identify copyright owners within the generated material. A multidimensional approach, combining policy, social, and technological solutions, proves more effective than isolated efforts. Langa [37] suggested for the labeling of AI-generated content to ensure transparency, while Ahmed et al. [38] emphasized the importance of proactive digital literacy campaigns to mitigate the negative impact of deepfakes. These combined strategies highlight the need for a comprehensive response to address the challenges posed by deepfake technology.

Table 7: Economic and Political Implications

Study	Aspect	Example	Impact
Islam et al. [36]	Fraudulent Transactions	AI-audio used to impersonate CEOs in financial scams.	Millions lost in fraudulent activities.
Efthymiou and Egleton [34]	Election Manipulation	Fake election speeches by politicians.	Erosion of public trust and instability of democracy.
Gilbert and Gilbert [32]	Financial Fraud	Fraudulent use of synthetic identity documentation.	Major economic and reputational harm to businesses.

Table 8: Detection and Countermeasure Strategies

Study	Strategy	Description	Challenges
Newman et al. [23]	AI-Based Detection	Frame-level anomaly detection in videos.	Adversarial AI evolving faster than detection tools.
Islam et al. [36]	Blockchain Verification	Verifying media authenticity through decentralized ledger systems.	High implementation cost and complexity.
Langa [37]	Content Labeling	Clear identification of AI-generated content for transparency.	Difficulty in standardizing labeling practices across platforms.
Ahmed et al. [38]	Public Education	Campaigns to educate users on recognizing AI-generated deception.	Reaching diverse and underserved populations.

6 Public Sentiment and Societal Perspectives of AI-Generated Contents

Understanding societal perspectives on AI-generated content is crucial, as public trust and acceptance significantly influence the social implications of misinformation and credibility. This section examines perceptions of credibility, the effects of misinformation campaigns, and cultural and demographic differences [39].

6.1 Perceptions of Credibility

Explicitly stating that content is AI-developed can significantly enhance its credibility, particularly when users are informed about the algorithmic tools used in its creation. Skepticism persists in scenarios where users cannot independently verify the authenticity of the content (as shown in Table 8), leading to a pessimistic outlook [40]. The lack of transparency in AI systems often results in distrust, especially when it comes to AI-generated disinformation [41]. Humans are generally perceived as more credible than AI, particularly in fields like journalism that rely heavily on emotional depth and human connection [42]. Additionally, demographic factors influence perceptions; for instance, Gen Z audiences are more receptive to AI-generated media when transparency markers are highlighted in Table 9, compared to older generations [43]. These findings underscore the importance of transparency and targeted communication strategies in fostering public trust in AI-generated content.

Table 9: Factors Influencing Perceptions of Credibility

Study	Factor	Impact	Example
Li et al. [40]	Content Labeling	Improves trust in labeled AI-generated content.	AI-generated weather reports with transparent labels.
Larsson and Heintz [41]	Transparency	Lack of transparency fosters distrust.	Unlabeled AI-generated news articles causing skepticism.
Huschens et al. [42]	Domain Sensitivity	Lower trust in sensitive areas like journalism and politics.	AI-written editorials are perceived as biased.
Ho et al. [43]	Generational Divide	Gen Z trusts labeled AI content more than older demographics.	Younger users favoring labeled AI-generated videos for educational content.

The use of AI-generated content for misinformation can turn out to be harmful for society as a whole. Deepfakes and AI-generated text are always used in the campaigns for disinformation, especially for political or social purposes. AI was used to sway voters during elections by spreading false information about candidates using deepfake videos [44]. Moreover, texts produced by AI are more believable because the models that produce them are sophisticated, and thus don't so many audience members struggle to differentiate between truth and lie (Table 10). AI-produced news articles

during a crisis tend to include both true and false elements which can lead to further confusion [45].

Table 10: Key Insights on AI-Driven Misinformation Campaigns

Study	Aspect	Description	Scenario
Williams et al. [45]	Electoral Manipulation	AI deepfakes influencing voter opinions during campaigns.	Fake videos of candidates endorsing false policies.
Chang et al. [44]	Amplification via Social Media	Rapid spread of AI misinformation on digital platforms.	Fake news about stock market crashes shared widely.

6.2 Cultural and Demographic Variations

Demographics, cultural contexts, and other factors significantly influence public perceptions regarding the acceptability of AI-generated content. Western consumers often exhibit greater skepticism toward content incorporating AI elements, viewing it with caution, whereas Asian markets tend to adopt a more optimistic perspective, often associating such content with technological advancement and progress [9, 46, 47]. Additionally, the perceived trustworthiness of AI-generated content is shaped by demographic variables, including education levels and digital literacy. According to Chu-Ke and Dong [48], highly educated individuals are more likely to critically evaluate AI-generated content, requiring a balanced understanding of its implications before acceptance. In contrast, individuals with lower levels of education may be more susceptible to misinformation, as they are less equipped to discern the credibility of such content .

6.3 AI-Generated Media Risks and Mitigation Strategies

The rise of AI-generated content has introduced significant ethical challenges, particularly concerning trust in media, public perception, and regulatory oversight [49]. As AI systems advance, their ability to produce hyper-realistic synthetic media—such as deepfake videos, fabricated news articles, and manipulated speeches—has grown exponentially. This has led to a decline in public trust in digital information, fostering widespread skepticism even at institutional levels [50]. Concerns over misinformation and manipulation have become pervasive, with younger, digitally savvy audiences struggling to verify the authenticity of AI-generated content. This tension highlights an ongoing conflict between technological innovation and the preservation of originality and trust in the media landscape [20], as represented in Figure 6.

The broader implications of generative AI research are critical to understanding its ethical dimensions. Initially celebrated for its potential in entertainment and creative industries, deepfake technology has increasingly been exploited for malicious purposes, including fraud, political propaganda, and social engineering attacks [31]. The socio-economic impacts are well-documented, with notable cases involving AI-generated voice scams and fabricated images of corporate executives, resulting in substantial

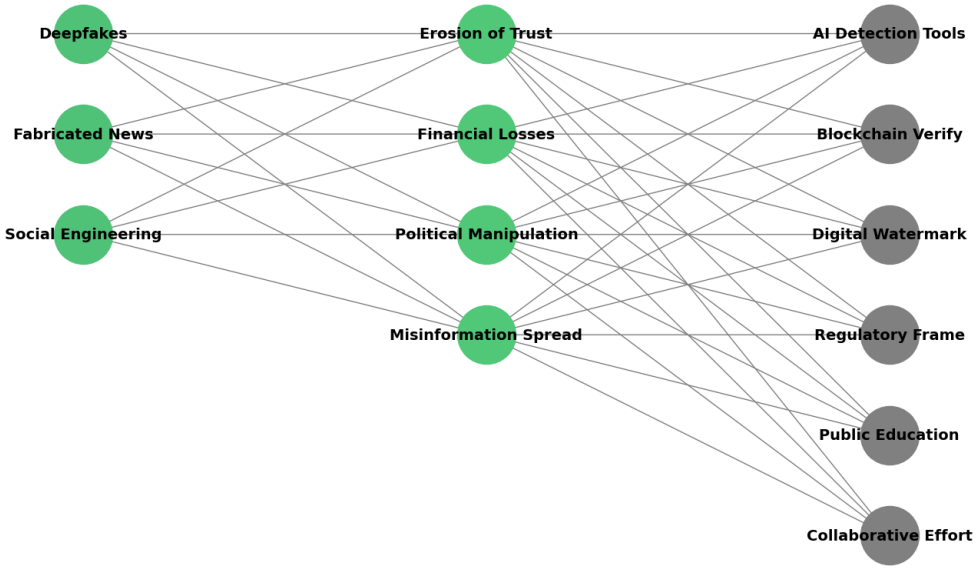

Fig. 6: Framework for Addressing AI-Generated Media Risks: Visualizing Risks, Impacts, and Mitigation Strategies to Combat Deepfakes and Misinformation

financial losses [36]. On a societal level, AI-driven misinformation campaigns have altered public perceptions and even influenced election outcomes, as fabricated narratives proliferate at an unprecedented scale [34]. While detection tools have improved, AI systems continue to outpace these countermeasures, rendering existing solutions less effective over time [23]. Addressing these challenges requires a balanced approach that integrates technological innovation, public education, and regulatory enforcement. Strategies such as content labeling and blockchain verification have been proposed to enhance transparency and accountability [36, 37]. Regulatory frameworks, such as the EU AI Act, underscore the growing recognition of AI governance as a global priority [27]. However, the rapid evolution of AI-generated media necessitates continuous advancements in both policy and detection technologies to safeguard digital ecosystems. Achieving sustainability in this domain demands collaboration among researchers, policymakers, and technology developers to address the ethical concerns of AI-generated content while fostering innovation and maintaining public trust in the digital age.

7 Comparative Analysis

This section provides comparative analysis of trust and deception in AI-generated media, cross-cultural differences and global case studies.

Table 11: Trust vs. Deception in AI Media

Study	Region	Trust Factors	Deception Factors
Wittenberg et al. [51]	Transparency	Clear labeling enhances credibility.	Lack of transparency erodes confidence.
ÓhÉigeartaigh et al. [52]	Cultural Context	Integration of cultural values fosters trust.	Contextual bias reduces global applicability.
Amoruso et al. [53]	Deepfakes	Credibility in non-controversial areas.	Misinformation spreads faster than detection tools.
Baldauf et al. [54]	Demographics	Younger users trust AI-driven solutions more readily.	Older populations are more skeptical.
Jaidka et al. [55]	AI Advancement	Ethical AI design fosters trust.	Rapid AI evolution complicates countermeasures.

The comparative analysis of trust and deception in AI-generated media highlights the interplay of transparency, cultural context, and demographic influences, as outlined in Table 11. Trust in AI media is strengthened through ethical design and transparency mechanisms, such as clear labeling, which improves credibility for non-controversial applications like weather forecasting [51]. Culturally tailored initiatives, such as Nordic public AI policies that integrate regional values, demonstrate context-specific trust-building strategies, though their efficacy varies across populations [52]. Demographically, younger, technology-literate individuals exhibit greater trust in AI-driven solutions compared to older, more skeptical groups [54]. Conversely, deception risks, particularly from deepfakes, undermine trust by facilitating the rapid spread of misinformation, as seen in politically charged contexts [53]. The accelerated evolution of AI complicates detection efforts, with malicious content often outpacing countermeasures [55]. This duality underscores the necessity of balancing ethical transparency with adaptive strategies to mitigate deception risks in AI-generated media.

7.1 Global Perspectives on Trust and Misinformation

In Western nations, trust in AI media is influenced by ethical concerns and privacy issues, with European countries emphasizing compliance with GDPR to build trust, though political misinformation remains a significant challenge [56]. In contrast, Asian nations such as China exhibit a more positive view of AI, particularly in relation to government approval and its integration into state media for mass communication, though state-driven bias can undermine objectivity [48]. Japan and South Korea are more liberal in their use of AI in newsrooms, valuing its accuracy and speed, although cultural acceptance can sometimes overshadow ethical concerns [57]. Developing nations, particularly in Southeast Asia and Africa, face infrastructural barriers that hinder trust in AI, especially among rural populations who are highly susceptible to AI-powered misinformation. Addressing these issues requires both educational initiatives and community engagement [55]. These varying regional perspectives on AI media are summarized in Table 12, which outlines the key trust and deception factors across different cultures, as illustrated in Figure 7.

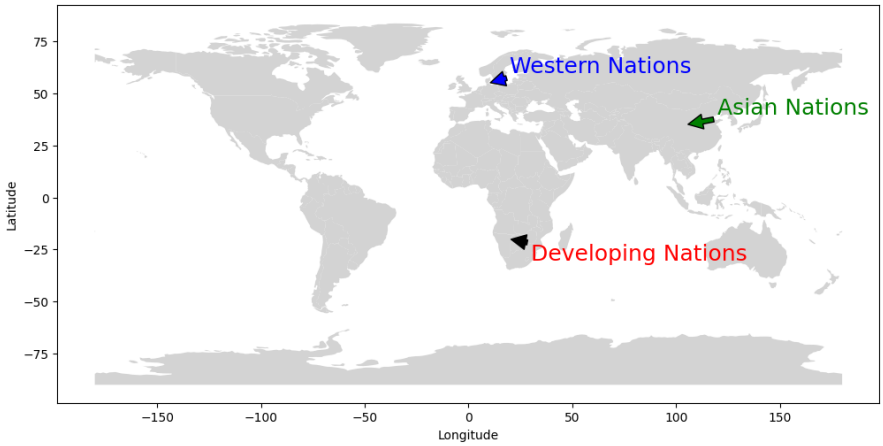

Fig. 7: Regional Insights on AI Media Trust and Misinformation: A Global Overview.

Table 12: Cross-Cultural Perceptions of AI Media

Study	Region	Trust Factors	Deception Factors
Lythe et al. [56]	Western Nations	GDPR compliance builds trust.	Misinformation damages democratic institutions.
Chu-Ke and Dong [48]	China	Trust linked to government approval.	State-driven bias undermines objectivity.
Al-Sharafi et al. [57]	Japan & South Korea	Efficiency and accuracy enhance public trust.	Cultural acceptance masks ethical concerns.
Jaidka et al. [55]	Developing Nations	Community programs improve AI literacy.	Rural populations face misinformation vulnerabilities.

7.2 Case Studies

The impact of AI technologies on various sectors has been profound, particularly in the areas of political campaigning [58], journalism, and combating misinformation. Each case highlights the distinct challenges and benefits that AI brings to these fields, demonstrating both the opportunities and risks associated with its adoption [59, 60]. This subsection presents three case studies that illustrate AI's role in shaping public perception, trust, and the efforts to combat misinformation. These case studies shed light on the multifaceted applications of AI (Table 13), its potential for positive change, and the need for robust countermeasures to mitigate its risks.

7.2.1 Case Study 1: Deepfake Dilemmas in Political Campaigns

In 2020, during the presidential elections in the USA, deepfake videos played a significant role in influencing voters' perceptions of political candidates. A study revealed that although many voters recognized these videos as altered [61], the deepfakes raised important concerns regarding the authenticity of political content. The phenomenon highlighted the need for surveillance systems to detect and prevent the spread of

manipulated media, drawing an analogy to the role of surveillance during a live scan to identify fakes [53].

7.2.2 Case Study 2: AI-Driven Journalism and Public Trust in Japan

In Japan, the Nihon Hōsō Kyōkai (NHK) network implemented AI technology to deliver weather reports, achieving an 85% satisfaction rate among viewers regarding the clarity of the reports. The use of AI in journalism, particularly in the context of weather forecasting, demonstrated how AI can enhance transparency and improve public trust in news delivery [62]. The NHK case emphasizes the importance of AI-driven transparency as a key factor in gaining public trust and satisfaction [57].

7.2.3 Case Study 3: COVID-19 Misinformation in Southeast Asia

Southeast Asia faced significant challenges with misinformation during the COVID-19 pandemic, especially in rural areas [63, 64]. A study demonstrated that misinformation campaigns disproportionately affected rural populations, further deepening the digital divide. However, educational campaigns tailored to local populations proved effective in combating the spread of AI-generated misinformation. These community-driven initiatives highlighted the critical role of education and grassroots efforts in addressing the negative impacts of AI in developing regions [55].

Table 13: Case Studies of AI Impact

Study	Region	Scenario	Impact	Counter Measures
Amoruso et al. [53]	United States	Deepfake videos in elections.	Eroded trust in political institutions.	Real-time deepfake detection systems.
Al-Sharafi et al. [57]	Japan	AI-driven journalism at NHK.	High public satisfaction with AI transparency.	Continued investment in ethical AI practices.
Jaidka et al. [55]	Southeast Asia	COVID-19 misinformation campaigns.	Rural populations disproportionately affected.	Community-driven education programs.

8 Discussion

The rapid development of AI technologies and their integration into the media sector has sparked significant discussions regarding the ethical, societal, and policy implications of AI-generated content. As AI continues to shape the media landscape, it brings forth complex questions surrounding authenticity, accountability, and the potential for misinformation. This section explores the broader implications of AI in media, examining how it affects media ethics, the role of AI developers in ensuring ethical standards [65], and the necessary societal and policy recommendations to guide the responsible use of AI in media production.

8.1 Ethical Implications of AI in Media

The integration of AI in media content creation presents crucial ethical challenges that require a revaluation of traditional media ethics in the context of the digital age [66]. The ability of AI to produce content indistinguishable from human-generated material raises concerns about authenticity and the risk of spreading misinformation [67]. Media organizations must balance the advantages of AI in content creation with the ethical responsibility of maintaining transparency and avoiding the blurring of boundaries between AI-generated and human-created content. Transparency becomes a central ethical principle, necessitating that media outlets inform their audiences when AI systems are involved in content production to preserve trust. Additionally, ethical guidelines must address not only the accountability of AI users but also the biases inherent in AI systems, which may perpetuate stereotypes or be misused to disseminate disinformation [68].

8.2 The Role of AI Developers in Ethical Media Practices

AI developers play a pivotal role in shaping the ethical future of AI-generated media. It is crucial for developers to design AI systems that adhere to ethical principles and consider the societal impacts of their applications [69]. While technical proficiency is important, developers must also ensure that AI systems are designed to prevent misuse, such as the creation of deepfakes or misleading content [31]. Collaboration with ethicists and social scientists is essential for developers to better understand the broader social implications of AI technologies and incorporate those insights into the design and deployment of these systems [70].

8.3 Policy and Societal Recommendations for AI Governance

Policy interventions surrounding AI in the media sector tend to be fragmented, making it difficult to address the comprehensive challenges posed by AI technologies. To effectively manage these issues, policies should prioritize the promotion of digital literacy, enabling citizens to navigate and critically evaluate AI-generated content responsibly. There is an urgent need for regulatory frameworks that mandate transparency in AI-generated content and ensure the responsible use of AI tools in media [71]. One potential solution is the implementation of "AI ethics audits," which could become mandatory regulatory processes to assess the ethical implications of AI technologies before their deployment [24]. Additionally, policies should encourage the development of AI technologies that align with democratic values and support public discourse, ensuring that AI serves as a tool that enhances, rather than detracts from, societal well-being [72].

9 Proposed Framework for Ethical AI-Generated Media

The rapid integration of AI into media ecosystems has introduced significant ethical, societal, and technical challenges, including trust erosion, misinformation, and cultural

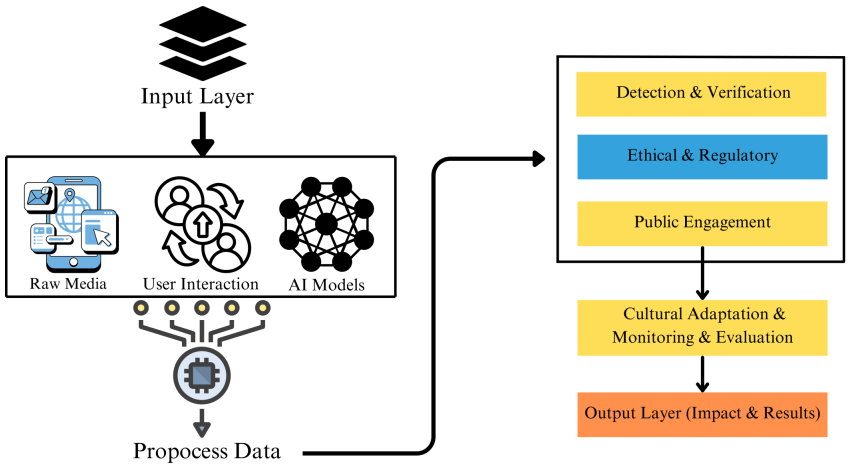

Fig. 8: Proposed framework for ethical AI-generated media, integrating detection, ethics, public engagement, cultural adaptation, and monitoring to protect trust in digital ecosystems.

disparities. To address these limitations and provide impactful solutions, a comprehensive framework is proposed. This framework integrates technological, regulatory, educational, and cultural dimensions to ensure ethical AI deployment, rebuild trust, and mitigate the risks associated with AI-generated media, as illustrated in Figure 8. The proposed AI-Mediated Trust Resilience Framework encompasses various layers and components, each contributing to a holistic solution to the challenges posed by AI in the media sector.

9.1 Stage 1: Technological Solutions

The first pillar of the proposed framework focuses on technological solutions, which are vital for ensuring that AI-generated content is both authentic and transparent. One of the most pressing concerns in AI-generated media is the spread of deepfakes and other synthetic content that can easily deceive audiences [73]. To address this, advanced detection systems are necessary. These systems would use AI-powered tools to detect anomalies in media, such as inconsistencies in videos or audio, providing early identification of manipulated content. Implementing frame-level anomaly detection models that achieve high accuracy rates (92%) will make it easier to identify deepfakes before they spread [74]. Moreover, blockchain-based verification systems [75] could offer a solution to authenticate digital content origins. By embedding invisible markers for copyright protection, blockchain ensures that media can be traced back to its original source, reducing the risk of content manipulation and misuse.

In addition to detection, transparency tools are essential to rebuild trust. One of the most significant challenges of AI-generated content is the lack of clarity on its origins. Mandating content labeling that clearly identifies when AI tools are used in media production would help users navigate AI-generated content more responsibly

[76]. Furthermore, explainable AI (XAI) systems would provide insight into the creation process of AI-generated content [77], ensuring that users can trace back how a piece of content was produced and understand the underlying algorithms. This accountability mechanism would not only foster trust but also deter unethical content creation. Integrating ethical AI design is essential to prevent AI systems from being misused. AI systems must be designed to prioritize fairness, accountability, and transparency, and collaborations with ethicist and social scientists will ensure these values are embedded from the ground up.

9.2 Stage 2: Regulatory and Policy Interventions

Global regulatory frameworks were established to ensure standardized governance across borders. Media outlets should be required to disclose the use of AI in content creation through transparency laws, which would mandate content labelling [78]. This regulation would establish boundaries on the ethical use of AI in the media industry and prevent misleading or malicious content. Additionally, adopting AI governance models, like the EU AI Act [79, 80], would help ensure compliance with ethical standards, data privacy regulations, and the fair use of AI technologies. Independent oversight committees could audit AI-generated content for accuracy, authenticity, and compliance with ethical standards. These committees would serve as neutral third parties, offering unbiased evaluations of AI-generated media. Moreover, penalties for malicious AI use (such as deepfake-driven fraud or election manipulation) must be enacted to ensure that any attempt to misuse AI for nefarious purposes is met with swift legal action [81].

Cross-border collaboration is another key aspect in addressing AI-driven misinformation globally [82, 83]. AI technologies do not recognize national boundaries, and therefore, global task forces must be formed to share best practices and coordinate efforts to detect and mitigate AI-generated deception. Such cross-border agreements would help address the geopolitical implications of AI-generated media, including election interference, which has already been shown to be a significant risk in several democratic processes worldwide.

9.3 Stage 3: Educational and Public Awareness Initiatives

Digital literacy programs are one of the first steps toward ensuring that the public can identify and critically evaluate AI-generated content [84, 85]. These programs would educate users about the risks of AI-generated misinformation and teach them how to verify content authenticity. By launching nationwide digital literacy campaigns [86], governments and organizations can promote awareness about the dangers of AI-driven media manipulation, particularly in an era where misinformation is increasingly pervasive. Additionally, developing educational modules in schools and universities focusing on AI ethics [87] and media literacy will prepare future generations to interact with AI in an informed and responsible manner.

Media literacy for vulnerable populations [88], particularly rural and under served communities, is an important step in protecting these groups from AI-driven misinformation. Community-driven education programs should be developed to address

the unique needs of these populations. By working with local organizations to deliver targeted workshops on identifying and combating misinformation, we can help these communities become more resilient against the spread of AI-generated deception [89].

10 Challenges and Limitations

The study identifies critical gaps in current research on AI-generated media, including the absence of integrated ethical frameworks addressing bias, transparency, and accountability. It highlights regional and cultural blind spots, with limited exploration of how public trust in synthetic media varies across demographics or regulatory landscapes. Additionally, long-term societal and economic impacts—such as the erosion of trust in democratic institutions—remain understudied, while interdisciplinary mitigation strategies combining policy, education, and technology are scarce. Geographic and linguistic biases further skew findings, as prior work disproportionately focuses on Western contexts and English-language studies, overlooking global perspectives.

To address these limitations, the study employs a systematic literature review (SLR) aligned with PRISMA guidelines, synthesizing 72 peer-reviewed articles (2015–2025) from 11 interdisciplinary databases. Content analysis identifies key themes like trust dynamics, detection vulnerabilities, and cultural disparities, supported by statistical insights (85% of users expressed concern about AI-generated misinformation in 2024). Bibliometric mapping reveals publication trends and institutional contributions, while bias mitigation strategies—such as multi-source data inclusion and dual independent reviewers—enhance methodological rigor. These approaches ensure a balanced, evidence-based synthesis of technical, cultural, and governance dimensions. The proposed AI-Mediated Trust Resilience Framework integrates technological solutions (anomaly detection, blockchain verification), policy interventions (transparency laws, cross-border collaboration), and educational initiatives (AI literacy programs).

11 Future Research Directions

The proposed framework highlights the necessity for future research to address the multifaceted challenges posed by AI-generated media. These challenges span ethical integration, technological advancements, longitudinal societal impacts, and regulatory strategies. By focusing on these areas, researchers can develop sustainable solutions that balance innovation with ethical accountability [90], ensuring the responsible use of AI in media ecosystems.

11.1 Ethical Integration of AI in Media Production

Future research must prioritize the development of comprehensive ethical frameworks that align AI-generated content with societal values. This requires interdisciplinary collaboration between ethicists, technologists, and media practitioners to address the dual-use nature of AI, where its creative potential is weighed against risks such as misinformation, bias, and trust erosion [87]. Studies should investigate how ethical

guidelines influence stakeholder perceptions of content credibility, particularly in sensitive domains like journalism and political communication [77]. For instance, the formulation of industry-wide standards for transparency in AI tools could mitigate public skepticism while fostering innovation [79]. Research should also explore the role of explainable AI (XAI) systems in providing insights into the decision-making processes of AI algorithms, ensuring accountability and trustworthiness in media production . By embedding ethical principles into AI design, future studies can help establish a foundation for responsible AI deployment that upholds societal values and public trust.

11.2 Enhancing AI Detection Systems

Advancing detection technologies is critical to counter the evolving threats posed by AI-driven deception, particularly deepfakes and synthetic media. Research should focus on improving the accuracy and robustness of detection tools, such as frame-level anomaly detection models, which have demonstrated high efficacy in identifying manipulated content [74]. However, the rapid evolution of adversarial techniques necessitates continuous innovation to stay ahead of malicious actors . Future studies must also evaluate the scalability and practicality of blockchain-based verification systems, which offer a decentralized approach to authenticating digital content origins and embedding invisible markers for copyright protection [75]. Additionally, the integration of digital watermarking techniques can enhance the traceability of AI-generated media, ensuring its authenticity and reducing the risk of misuse [81]. Real-world trials and simulations are essential to identify implementation challenges, such as computational costs, interoperability with existing systems, and the adaptability of detection tools across diverse media ecosystems [73]. By addressing these challenges, researchers can develop more resilient and efficient systems to combat AI-driven deception.

11.3 Longitudinal Impact Studies on Public Perception

Longitudinal studies are essential to understand the long-term effects of AI-generated content on public trust and behavior. Research should track shifts in public perception across different demographics and regions, analyzing how factors such as age, education, and cultural context influence trust in AI-generated media [86, 88]. For instance, younger generations, such as Gen Z, may exhibit higher trust in labeled AI content compared to older populations, highlighting the need for tailored communication strategies [84]. Longitudinal research can also assess the efficacy of digital literacy campaigns in improving resilience against misinformation, particularly in vulnerable populations such as rural communities [85]. By monitoring changes in public perception over time, researchers can provide actionable insights for media producers to adapt their content strategies, ensuring credibility and trust in an increasingly AI-driven media landscape [77]. These studies will also help identify emerging trends and challenges, enabling proactive measures to address evolving societal concerns.

11.4 Regulatory and Educational Strategies for AI Media

Future research must evaluate the effectiveness of regulatory frameworks in enforcing transparency, accountability, and ethical standards in AI-generated media. Comparative analyses of regional policies, such as the EU AI Act [79], can identify best practices for global governance, particularly in mitigating risks like election interference and misinformation [83]. Studies should also investigate the role of independent oversight committees in auditing AI-generated content, ensuring its accuracy and compliance with ethical standards [78]. On the educational front, research should focus on the development and implementation of tailored programs to enhance AI literacy among diverse populations. Community-driven workshops and digital literacy campaigns can empower vulnerable groups, such as rural and underserved communities, to critically evaluate AI-generated content and resist misinformation [76, 89]. By combining regulatory and educational strategies, future research can create a holistic approach to mitigating the risks of AI-generated media while fostering public trust and engagement. These future research directions underscore the need for interdisciplinary collaboration to address the dual-use nature of AI, ensuring that technological innovation aligns with ethical accountability and societal trust [78, 82]. By focusing on these areas, researchers can contribute to the development of sustainable solutions that safeguard the integrity of digital ecosystems while promoting the responsible use of AI in media production.

12 Conclusion

The rapid integration of AI-generated media has brought transformative changes to content creation while raising significant ethical concerns. This study provides a comprehensive review of the ethical implications of AI in media ecosystems, focusing on the erosion of trust, technological vulnerabilities that enable deception, and cultural disparities in public perception. By analyzing 72 peer-reviewed studies from 2015 to 2025, the research highlights the need to balance technological advancements with ethical considerations to safeguard democratic integrity and societal trust. Additionally, the study proposes the AI-Mediated Trust Resilience Framework, which offers a multidisciplinary approach to integrating detection technologies, policy reforms, and public education to address these challenges. The research also identifies future research opportunities in four key domains: Ethical Integration of AI in Media Production, Enhancing AI Detection Systems, Longitudinal Impact Studies on Public Perception, and Regulatory and Educational Strategies for AI Media. The research directly addresses three key research questions.

- **RQ-1:**Our review identifies that AI-generated content—particularly deepfakes and synthetic text—exacerbates trust erosion by blurring authenticity and enabling misinformation. Mechanisms like algorithmic opacity (undisclosed AI use in journalism) and malicious campaigns (political deepfakes) degrade public confidence. However, transparency measures (e.g., content labeling) and AI-powered detection tools (frame-level anomaly detection) emerge as critical mitigators, restoring credibility when implemented rigorously.

- **RQ-2:** The analysis reveals vulnerabilities in detection systems, such as the inability of current tools to keep pace with adversarial AI (hybrid synthetic media combining text, audio, and visuals). Sociocultural drivers, including political propaganda and economic fraud (CEO impersonation scams), exploit these gaps. The study advocates for adaptive detection frameworks (blockchain verification) and policy interventions (stricter content-labeling laws) to curb malicious use.
- **RQ3:** Cultural, demographic, and regional differences shape public perception of synthetic media. Younger, digitally literate populations demonstrate higher trust in labeled AI content, while regions with limited infrastructure face heightened risks due to low media literacy. Cultural contexts further influence acceptance, with disparities between skepticism in Western contexts and optimism in Asian markets.

This research underscores the need for a comprehensive, interdisciplinary approach to mitigate the ethical challenges posed by AI-generated media. The proposed AI-Mediated Trust Resilience Framework provides actionable solutions to restore trust and accountability in digital ecosystems. Despite limitations such as geographic bias and the rapid pace of AI developments. As AI technologies continue to evolve rapidly, it is crucial that innovation in AI-generated media be harmonized with ethical accountability to prevent the erosion of trust and ensure the preservation of democratic values. The future of AI in media relies not just on technical advancements, but on responsible, transparent, and culturally aware implementations that prioritize societal integrity.

Funding

This research is funded by Woosong University Academic Research 2025.

Conflict of Interest

There are no conflicts of interest associated with this work.

References

[1] Maia, C., Ariel, P., Nunes, S.: Adding human values on the deepfake: co-designing fact-checking solutions to combat misinformation. AI and Ethics, 1–16 (2024) https://doi.org/10.1007/s43681-024-00619-y

[2] Al-kfairy, M., Mustafa, D., Kshetri, N., Insiew, M., Alfandi, O.: Ethical challenges and solutions of generative ai: An interdisciplinary perspective. Informatics **11**(3), 58 (2024) https://doi.org/10.3390/informatics11030058

[3] Karaş, Z.: Effects of ai-generated misinformation and disinformation on the economy. Düzce Üniversitesi Bilim ve Teknoloji Dergisi **12**(4), 2349–2360 (2024) https://doi.org/10.29130/dubited.1537268

[4] Hate, C.f.C.D.: The double-edged sword of ai: How generative language models like google bard and chatgpt pose a threat to countering hate and misinformation

online. Harvard Data Science Review (Special Issue 5) (2024) https://doi.org/10.1162/99608f92.be4e28f0

[5] Ghiurău, D., Popescu, D.E.: Distinguishing reality from ai: Approaches for detecting synthetic content. Computers **14**(1), 1 (2024) https://doi.org/10.3390/computers14010001

[6] Ku, W., Liu, Y., Zhang, W., An, P.: Genpod: Constructive news framing in ai-generated podcasts more effectively reduces negative emotions than non-constructive framing. arXiv preprint (2024) https://doi.org/10.48550/arXiv.2412.18300

[7] Kreps, S., Kriner, D.: How ai threatens democracy. Journal of Democracy **34**(4), 122–131 (2023) https://doi.org/10.1353/jod.2023.a907693

[8] Aleessawi, N., Alzubi, S.: The implications of artificial intelligence (ai) on the quality of media content. Studies in Media and Communication **12**(4), 41–51 (2024) https://doi.org/10.11114/smc.v12i4.7058

[9] Frank, J., Herbert, F., Ricker, J., Schönherr, L., Eisenhofer, T., Fischer, A., Dürmuth, M., Holz, T.: A representative study on human detection of artificially generated media across countries. In: 2024 IEEE Symposium on Security and Privacy (SP), pp. 55–73. IEEE, San Francisco, CA, USA at the Hilton San Francisco Union Square (2024). https://doi.org/10.48550/arXiv.2312.05976

[10] Ranade, N., Saravia, M.: Teaching ai ethics in technical and professional communication: A systematic review. IEEE Transactions on Professional Communication **67**(4), 422–436 (2024) https://doi.org/10.1109/TPC.2024.3458708

[11] Contreras, M.R., Jaimes, J.O.P.: Ai ethics in the fields of education and research: A systematic literature review. In: 2024 International Symposium on Accreditation of Engineering and Computing Education (ICACIT), pp. 1–6 (2024). https://doi.org/10.1109/ICACIT62963.2024.10788651

[12] Xu, Q., Gu, J., Lu, J.: Leveraging artificial intelligence and large language models for enhanced teaching and learning: A systematic literature review. In: 2024 13th International Conference on Computer Technologies and Development (TechDev), pp. 73–77 (2024). https://doi.org/10.1109/TechDev64369.2024.00021

[13] Gunawan, A., Wiputra, R.: Ethical implications of generative ai in medicine: A systematic review of current practices and future directions. In: 2024 IEEE 12th Conference on Systems, Process & Control (ICSPC), pp. 239–244 (2024). https://doi.org/10.1109/ICSPC63060.2024.10862569

[14] Masciari, E., Umair, A., Ullah, M.H.: A systematic literature review on ai-based recommendation systems and their ethical considerations. IEEE Access **12**, 121223–121241 (2024) https://doi.org/10.1109/ACCESS.2024.3451054

[15] Bena, Y.A., Ibrahim, R., Mahmood, J., Al-Dhaqm, A., Alshammari, A., Nasser, M., Nura Yusuf, M., Ayemowa, M.O.: Big data governance challenges arising from data generated by intelligent systems technologies: A systematic literature review. IEEE Access **13**, 12859–12888 (2025) https://doi.org/10.1109/ACCESS.2025.3528941

[16] Han, Y., Hou, J., Sun, Y.: Research and application of gpt-based large language models in business and economics: A systematic literature review in progress. In: 2023 IEEE International Conference on Computing (ICOCO), pp. 118–123 (2023). https://doi.org/10.1109/ICOCO59262.2023.10397642

[17] Abokhoza, R., Al-Tkhayneh, K.M., Al Qaruty, R., Abdel Hadi, S., Ellala, Z.K.: The ethics of applying artificial intelligence (ai) for communication governance. In: 2024 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS), pp. 236–241 (2024). https://doi.org/10.1109/ICCNS62192.2024.10776312

[18] Wiese, L.J., Schiff, D.S., Magana, A.J.: Being proactive for responsible ai: Analyzing multiple sectors for innovation via systematic literature review. In: 2023 IEEE International Symposium on Ethics in Engineering, Science, and Technology (ETHICS), pp. 1–1 (2023). https://doi.org/10.1109/ETHICS57328.2023.10154947

[19] Koralage, M., Malagala, G., Wijemunige, N., De Silva, S., Banda, W., Lokeshwara, A.: From chalkboards to chatbots: A systematic literature review on the evolution of ai in accounting pedagogy. In: 2024 6th International Conference on Advancements in Computing (ICAC), pp. 61–66 (2024). https://doi.org/10.1109/ICAC64487.2024.10851039

[20] Fernandez, M., Alani, H.: Online misinformation: Challenges and future directions. In: Companion Proceedings of the the Web Conference 2018, pp. 595–602 (2018). https://doi.org/10.1145/3184558.3188730

[21] Krüger, S., Wilson, C.: The problem with trust: on the discursive commodification of trust in ai. AI & SOCIETY **38**(4), 1753–1761 (2023) https://doi.org/10.1007/s00146-022-01401-6

[22] Jiang, X., Wu, Z., Yu, F., *et al.*: Constructing consumer trust through artificial intelligence generated content. Academic Journal of Business & Management **6**(8), 263–272 (2024) https://doi.org/10.25236/AJBM.2024.060839

[23] Newman, N., Fletcher, R., Schulz, A., Andi, S., Robertson, C., Nielsen, R.: Reuters Institute digital news report 2021. Reuters Institute for the Study of Journalism. ISBN: 9781914566158 (2021). https://ora.ox.ac.uk/objects/uuid:219692c0-85ce-4cab-9cbc-d3cdffabf62b

[24] Jobin, A., Ienca, M., Vayena, E.: The global landscape of ai ethics guidelines. Nature machine intelligence **1**(9), 389–399 (2019) https://doi.org/10.1038/s42256-019-0088-2

[25] Fletcher, R., Nielsen, R.: Are people incidentally exposed to news on social media? a comparative analysis. New media & society **20**(7), 2450–2468 (2018) https://doi.org/10.1177/1461444817724170

[26] Nguyen, D.C., Ding, M., Pathirana, P.N., Seneviratne, A.: Blockchain and ai-based solutions to combat coronavirus (covid-19)-like epidemics: A survey. IEEE Access **9**, 95730–95753 (2021) https://doi.org/10.1109/ACCESS.2021.3093633

[27] Mondal, H., Mondal, S.: Artificial intelligence-generated content needs a human oversight. Indian Journal of Dermatology **69**(3), 284 (2024) https://doi.org/10.4103/ijd.ijd_163_24

[28] Brinkerhoff, D.W., Brinkerhoff, J.M.: Public sector management reform in developing countries: Perspectives beyond npm orthodoxy. Public Administration and Development **35**(4), 222–237 (2015) https://doi.org/10.1002/pad.1739

[29] Nader, K., Toprac, P., Scott, S., Baker, S.: Public understanding of artificial intelligence through entertainment media. AI & society **39**(2), 713–726 (2024) https://doi.org/10.1007/s00146-022-01427-w

[30] Kolo, C., Mütterlein, J., Schmid, S.A.: Believing journalists, ai, or fake news: The role of trust in media. In: HICSS, pp. 1–10 (2022). https://doi.org/10.24251/HICSS.2022.393

[31] Heidari, A., Jafari Navimipour, N., Dag, H., Unal, M.: Deepfake detection using deep learning methods: A systematic and comprehensive review. Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery **14**(2), 1520 (2024) https://doi.org/10.1002/widm.1520

[32] Gilbert, C., Gilbert, M.A.: The role of artificial intelligence (ai) in combatting deepfakes and digital misinformation. International Research Journal of Advanced Engineering and Science (ISSN: 2455-9024) **9**(4), 170–181 (2024) https://doi.org/10.2139/ssrn.5165104

[33] Steingartner, W., Galinec, D.: Cyber threats and cyber deception in hybrid warfare. Acta Polytechnica Hungarica **18**(3), 25–45 (2021) https://doi.org/10.12700/APH.18.3.2021.3.2

[34] Efthymiou, I.-P., Egleton, T.E.: Artificial intelligence for sustainable smart cities. In: Handbook of Research on Applications of AI, Digital Twin, and Internet of Things for Sustainable Development. IGI Global, Hershey, PA, USA (2023). https://doi.org/10.4018/978-1-6684-6821-0.ch001

[35] Diakopoulos, N.: Accountability in algorithmic decision making. Communications of the ACM **59**(2), 56–62 (2016) https://doi.org/10.1145/2844110

[36] Islam, M.B.E., Haseeb, M., Batool, H., Ahtasham, N., Muhammad, Z.: Ai threats to politics, elections, and democracy: a blockchain-based deepfake authenticity verification framework. Blockchains **2**(4), 458–481 (2024) https://doi.org/10.3390/blockchains2040020

[37] Langa, J.: Deepfakes, real consequences: Crafting legislation to combat threats posed by deepfakes. BUL Rev. **101**, 761 (2021) https://doi.org/10.2139/ssrn.5165104

[38] Ahmed, S.R., Sonuç, E., Ahmed, M.R., Duru, A.D.: Analysis survey on deepfake detection and recognition with convolutional neural networks. In: 2022 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA), pp. 1–7 (2022). https://doi.org/10.1109/HORA55278.2022.9799858

[39] Merenkov, A.V., Campa, R., Dronishinets, N.: Public opinion on artificial intelligence development (2021) https://doi.org/10.1093/oxfordhb/9780197579329.013.36

[40] Li, F., Yang, Y., *et al.*: Impact of artificial intelligence–generated content labels on perceived accuracy, message credibility, and sharing intentions for misinformation: Web-based, randomized, controlled experiment. JMIR Formative Research **8**(1), 60024 (2024) https://doi.org/10.2196/60024

[41] Larsson, S., Heintz, F.: Transparency in artificial intelligence. Internet policy review **9**(2) (2020) https://doi.org/10.14763/2020.2.1469

[42] Huschens, M., Briesch, M., Sobania, D., Rothlauf, F.: Do you trust chatgpt?–perceived credibility of human and ai-generated content. arXiv preprint arXiv:2309.02524 (2023) https://doi.org/10.48550/arXiv.2309.02524

[43] Ho, M.-T., Mantello, P., Ghotbi, N., Nguyen, M.-H., Nguyen, H.-K.T., Vuong, Q.-H.: Rethinking technological acceptance in the age of emotional ai: Surveying gen z (zoomer) attitudes toward non-conscious data collection. Technology in Society **70**, 102011 (2022) https://doi.org/10.1016/j.techsoc.2022.102011

[44] Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H., Yi, X., Wang, C., Wang, Y., *et al.*: A survey on evaluation of large language models. ACM Transactions on Intelligent Systems and Technology **15**(3), 1–45 (2024) https://doi.org/10.48550/arXiv.2307.03109

[45] Williams, T., Ioannou, M., Tzani, C.: Artificially disinformed and radicalised: How ai produced disinformation could encourage radicalisation. Assessment and Development Matters **16**(1), 29–34 (2024) https://doi.org/10.53841/bpsadm.

2024.16.1.29

[46] Kaya, F., Aydin, F., Schepman, A., Rodway, P., Yetişensoy, O., and, M.D.K.: The roles of personality traits, ai anxiety, and demographic factors in attitudes toward artificial intelligence. International Journal of Human–Computer Interaction **40**(2), 497–514 (2024) https://doi.org/10.1080/10447318.2022.2151730

[47] Amiruzzaman, M., Zhao, Y., Amiruzzaman, S., Karpinski, A.C., Wu, T.H.: An ai-based framework for studying visual diversity of urban neighborhoods and its relationship with socio-demographic variables. Journal of Computational Social Science **6**(1), 315–337 (2023) https://doi.org/10.1007/s42001-022-00197-1

[48] Chu-Ke, C., Dong, Y.: Misinformation and literacies in the era of generative artificial intelligence: A brief overview and a call for future research. Emerging Media **2**(1), 70–85 (2024) https://doi.org/10.1177/27523543241240285

[49] Lyu, S.: Deepfake the menace: mitigating the negative impacts of ai-generated content. Organizational Cybersecurity Journal: Practice, Process and People (ahead-of-print) (2024) https://doi.org/10.1108/OCJ-08-2022-0014

[50] Shrungare, J.: Exploring synthetic media: How ai is changing art. XRDS **29**(4), 64–65 (2023) https://doi.org/10.1145/3596936

[51] Wittenberg, C., Epstein, Z., Berinsky, A.J., Rand, D.G.: Labeling AI-Generated Content: Promises, Perils, and Future Directions. An MIT Exploration of Generative AI (2024) https://doi.org/10.21428/e4baedd9.0319e3a6

[52] ÓhÉigeartaigh, S.S., Whittlestone, J., Liu, Y., Zeng, Y., Liu, Z.: Overcoming barriers to cross-cultural cooperation in ai ethics and governance. Philosophy & technology **33**, 571–593 (2020) https://doi.org/10.1007/s13347-020-00402-x

[53] Amoruso, M., Anello, D., Auletta, V., Cerulli, R., Ferraioli, D., Raiconi, A.: Contrasting the spread of misinformation in online social networks. Journal of Artificial Intelligence Research **69**, 847–879 (2020) https://doi.org/10.1613/jair.1.11509

[54] Baldauf, M., Fröehlich, P., Endl, R.: Trust me, i'ma doctor–user perceptions of ai-driven apps for mobile health diagnosis. In: Proceedings of the 19th International Conference on Mobile and Ubiquitous Multimedia, pp. 167–178 (2020). https://doi.org/10.1145/3428361.3428362

[55] Jaidka, K., Chen, T., Chesterman, S., Hsu, W., Kan, M.-Y., Kankanhalli, M., Lee, M.L., Seres, G., Sim, T., Taeihagh, A., *et al.*: Misinformation, disinformation, and generative ai: Implications for perception and policy. Digital Government: Research and Practice (2024) https://doi.org/10.1145/3689372

[56] Lythe, M., Cos, G.M., Mingallon, M., Lensen, A., Galloway, C., Knox, D., Auvaa,

S., Kumarasinghe, K.: Explainable ai–building trust through understanding (2023) https://doi.org/10.25455/wgtn.24531433

[57] Al-Sharafi, M.A., Al-Emran, M., Arpaci, I., Iahad, N.A., AlQudah, A.A., Iranmanesh, M., Al-Qaysi, N.: Generation z use of artificial intelligence products and its impact on environmental sustainability: A cross-cultural comparison. Computers in Human Behavior **143**, 107708 (2023) https://doi.org/10.1016/j.chb.2023.107708

[58] Lundberg, E.: The level of tourism development and resident attitudes: A comparative case study of coastal destinations. Scandinavian Journal of Hospitality and Tourism **15**(3), 266–294 (2015) https://doi.org/10.1080/15022250.2015.1005335

[59] Espín-León, A., Jimeno-Morenilla, A., Pertegal-Felices, M.L., Azorín-López, J.: Quantification of cultural identity through artificial intelligence: a case study on the waorani amazonian ethnicity. Soft Computing **24**(15), 11045–11057 (2020) https://doi.org/10.1007/s00500-019-04469-3

[60] Ahrweiler, P., Abe, J., Neumann, M.: Using a case study approach for investigating the status quo and future options of ai-based social. Participatory Artificial Intelligence in Public Social Services: From Bias to Fairness in Assessing Beneficiaries, 1 https://doi.org/10.1007/978-3-031-71678-2_1

[61] Campo-Ruiz, I.: Artificial intelligence may affect diversity: architecture and cultural context reflected through chatgpt, midjourney, and google maps. Humanities and Social Sciences Communications **12**(1), 1–13 (2025) https://doi.org/10.1057/s41599-024-03968-5

[62] Abulkassova, D., Muldasheva, G., Nurtazin, M., Tleukhanov, N., Kuspanova, A.: The phenomenon of artificial intelligence in modern transformational sociocultural processes: Socio-philosophical analysis. AI & SOCIETY, 1–11 (2025) https://doi.org/10.1007/s00146-025-02195-z

[63] Cho, C.S.: Illiberal responses to "fake news" in southeast asia. Democratization, 1–21 (2025) https://doi.org/10.1080/13510347.2024.2442395

[64] Suarez-Lledo, V., Ortega-Martin, E., Carretero-Bravo, J., Ramos-Fiol, B., Alvarez-Galvez, J., *et al.*: Unraveling the use of disinformation hashtags by social bots during the covid-19 pandemic: Social networks analysis. JMIR infodemiology **5**(1), 50021 (2025) https://doi.org/10.2196/50021

[65] Shao, Y., Yin, Q.: A review the role of artificial intelligence in media content creation for sdgs development. Journal of Lifestyle and SDGs Review **5**(2), 03971–03971 (2025) https://doi.org/10.47172/2965-730X.SDGsReview.v5.n02.pe03971

[66] Vrabič Dežman, D.: Promising the future, encoding the past: Ai hype and public media imagery. AI and Ethics **4**(3), 743–756 (2024) https://doi.org/10.1007/

s43681-024-00474-x

[67] Rodriguez, C.A., Gopalakrishnan, L., Del Cid, M., Folk, J.B., Yonek, J., Tolou-Shams, M.: The ethical implications of using social media to engage and retain justice-involved youth in behavioral health research. Journal of Empirical Research on Human Research Ethics **16**(4), 356–363 (2021) https://doi.org/10.1177/15562646211039701

[68] Saura, J.R., Palacios-Marqués, D., Iturricha-Fernández, A.: Ethical design in social media: Assessing the main performance measurements of user online behavior modification. Journal of Business Research **129**, 271–281 (2021) https://doi.org/10.1016/j.jbusres.2021.03.001

[69] Bingley, W.J., Curtis, C., Lockey, S., Bialkowski, A., Gillespie, N., Haslam, S.A., Ko, R.K., Steffens, N., Wiles, J., Worthy, P.: Where is the human in human-centered ai? insights from developer priorities and user experiences. Computers in Human Behavior **141**, 107617 (2023) https://doi.org/10.1016/j.chb.2022.107617

[70] Kazemitabaar, M., Chow, J., Ma, C.K.T., Ericson, B.J., Weintrop, D., Grossman, T.: Studying the effect of ai code generators on supporting novice learners in introductory programming. In: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, pp. 1–23 (2023). https://doi.org/10.1145/3544548.358091

[71] Floridi, L., Cowls, J.: A unified framework of five principles for ai in society. Machine learning and the city: Applications in architecture and urban design, 535–545 (2022) https://doi.org/10.1162/99608f92.8cd550d1

[72] Cowls, J., Tsamados, A., Taddeo, M., Floridi, L.: The ai gambit: leveraging artificial intelligence to combat climate change—opportunities, challenges, and recommendations. Ai & Society, 1–25 (2023) https://doi.org/10.1007/s00146-021-01294-x

[73] Love, P.E.D., Matthews, J., Fang, W., Porter, S., Luo, H., Ding, L.: Learning to comprehend and trust artificial intelligence outcomes: A conceptual explainable ai evaluation framework. IEEE Engineering Management Review **52**(1), 230–247 (2024) https://doi.org/10.1109/EMR.2023.3342200

[74] Klarić, M.: Regulation of ai technology implementation in public administration. In: 2024 47th MIPRO ICT and Electronics Convention (MIPRO), pp. 1450–1456 (2024). https://doi.org/10.1109/MIPRO60963.2024.10569836

[75] Al Ahmed, S., Mamun Rudro, R.A., Prity, A.J., Saha, S., Mansoor, N., Nur, K.: Credchain: Academic and professional certificate verification system using blockchain. In: 2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), pp. 1–6 (2024). https://doi.org/10.1109/iCACCESS61735.2024.10499520

[76] Oki, A., Ogawa, Y., Ota, K., Dong, M.: Evaluation of applying federated learning to distributed intrusion detection systems through explainable ai. IEEE Networking Letters **6**(3), 198–202 (2024) https://doi.org/10.1109/LNET.2024.3465516

[77] Sajid Farooq, M., Hassan Ghulam Muhammad, M., Ali, O., Zeeshan, Z., Saleem, M., Ahmad, M., Abbas, S., Khan, M.A., Ghazal, T.M.: Developing a transparent anaemia prediction model empowered with explainable artificial intelligence. IEEE Access **13**, 1307–1318 (2025) https://doi.org/10.1109/ACCESS.2024.3522080

[78] Wang, J., Huo, Y., Mahe, J., Ge, Z., Liu, Z., Wang, W., Zhang, L.: Developing an ethical regulatory framework for artificial intelligence: Integrating systematic review, thematic analysis, and multidisciplinary theories. IEEE Access **12**, 179383–179395 (2024) https://doi.org/10.1109/ACCESS.2024.3501332

[79] Stettinger, G., Weissensteiner, P., Khastgir, S.: Trustworthiness assurance assessment for high-risk ai-based systems. IEEE Access **12**, 22718–22745 (2024) https://doi.org/10.1109/ACCESS.2024.3364387

[80] Costa, H., Mendonça, J.: The proposal of an ai policy maturity model. In: 2024 IEEE Conference on Artificial Intelligence (CAI), pp. 1408–1413 (2024). https://doi.org/10.1109/CAI59869.2024.00251

[81] Park, C.: Roles of scenario planning for public policy: A case study of korean artificial intelligence policy. In: 2022 IEEE Technology and Engineering Management Conference (TEMSCON EUROPE), pp. 56–60 (2022). https://doi.org/10.1109/TEMSCONEUROPE54743.2022.9802050

[82] Saha, A., Sati, M.M., Naval, N., Sebastian, T., Jayamala, C., Sudharma, B.P.: Policy suggestions using artificial intelligence techniques – a machine learning algorithms approach. In: 2024 Second International Conference Computational and Characterization Techniques in Engineering & Sciences (IC3TES), pp. 1–6 (2024). https://doi.org/10.1109/IC3TES62412.2024.10877560

[83] Amicone, A., Marangoni, L., Marceddu, A., Miccoli, M.: Ai-based public policy making: A new holistic, integrated and "ai by design" approach. In: 2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), pp. 525–532 (2023). https://doi.org/10.1109/DCOSS-IoT58021.2023.00087

[84] Wang, D., Chen, G.: Making ai accessible for stem teachers: Using explainable ai for unpacking classroom discourse analysis. IEEE Transactions on Education **67**(6), 907–918 (2024) https://doi.org/10.1109/TE.2024.3421606

[85] Chiu, T.K.F., Meng, H., Chai, C.-S., King, I., Wong, S., Yam, Y.: Creation and

evaluation of a pretertiary artificial intelligence (ai) curriculum. IEEE Transactions on Education **65**(1), 30–39 (2022) https://doi.org/10.1109/TE.2021.3085878

[86] Lang, Q., Wang, M., Yin, M., Liang, S., Song, W.: Transforming education with generative ai (gai): Key insights and future prospects. IEEE Transactions on Learning Technologies **18**, 230–242 (2025) https://doi.org/10.1109/TLT.2025.3537618

[87] Kim, D., Zhu, Q., Eldardiry, H.: Toward a policy approach to normative artificial intelligence governance: Implications for ai ethics education. IEEE Transactions on Technology and Society **5**(3), 325–333 (2024) https://doi.org/10.1109/TTS.2024.3430940

[88] Almogren, A.S., Al-Rahmi, W.M., Dahri, N.A.: Integrated technological approaches to academic success: Mobile learning, social media, and ai in visual art education. IEEE Access **12**, 175391–175413 (2024) https://doi.org/10.1109/ACCESS.2024.3498047

[89] Chen, L., Chen, P., Lin, Z.: Artificial intelligence in education: A review. IEEE Access **8**, 75264–75278 (2020) https://doi.org/10.1109/ACCESS.2020.2988510

[90] Bennett, M.T., Maruyama, Y.: Philosophical specification of empathetic ethical artificial intelligence. IEEE Transactions on Cognitive and Developmental Systems **14**(2), 292–300 (2022) https://doi.org/10.1109/TCDS.2021.3099945